

Day 3

5 Data Types

int → whole number

age = 25

big = 1,000,000 → Python ignores ','

type(age) → <class 'int'>

Float → decimal numbers

price = 9.99

science = 1.5e4

prints 15000.0

str → text

name = "Prakash"

"42"

" " & ' ' (same)

""" """ (multiline)

bool → only two value

True

↑

False

↓

bool is secretly int

True == 1

False == 0

None → absence of value

result = None

type(None) → <class 'NoneType'>

Tools

type() → tells u what type of sth is

type(25)

→ int

"hi"

→ str

type(True)

→ bool

isinstance() → asks "is this THIS type?" returns T/F
 is (25, int) → T
 is (True, int) → T → bool subclass of int

Conversion

int("42") → 42 bool(0) → False
 int(3.999) → 3 bool("hi") → T
 int(True) → 1 " ("") → F
 float(42) → 42.0 " (None) → F
 str(100) → "100"
 bool(1) → True

Immutable & Mutable

int, float, str, bool, None, tuple

New obj created on every change

list, dict, set

Modified in place, affected

a = "hello"

b = a → points to same str

a = "world" → creates new, points a there

print(a) → world

"(b) → hello ← safe

"(a is b) → false ← diff obj now

a → "hello" (1111)

b → "hello" (1111)

a = [1, 2, 3]

b = a

b.append(4)

print(a) [1, 2, 3, 4]

"(b) [1, 2, 3, 4]

"(a is b) true ← safe

id() → returns memory address of object if changed location of obj

a → "world" (222)

b → "hello" (111)